UNIVERSITI MALAYA

NFMN

SIM 1005 / SIN 1002

Fundamentals of Spreadsheets / Introduction to Worksheet

Lecture 3: Graphical Tools in Excel

By

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Institute of Mathematical Sciences

Objectives

- To classify types of graphical tools in Excel.
- To use graphics and drawing tools.
- To create charts in Excel.
- To find the trend of a series of data.
- To find the best curve-fit model for a series of data.

Original notes by NFM Noor

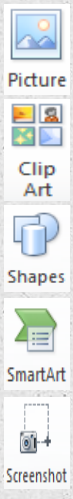
Graphical Tools in Excel

- Graphics
- Drawing Tools
- Charts

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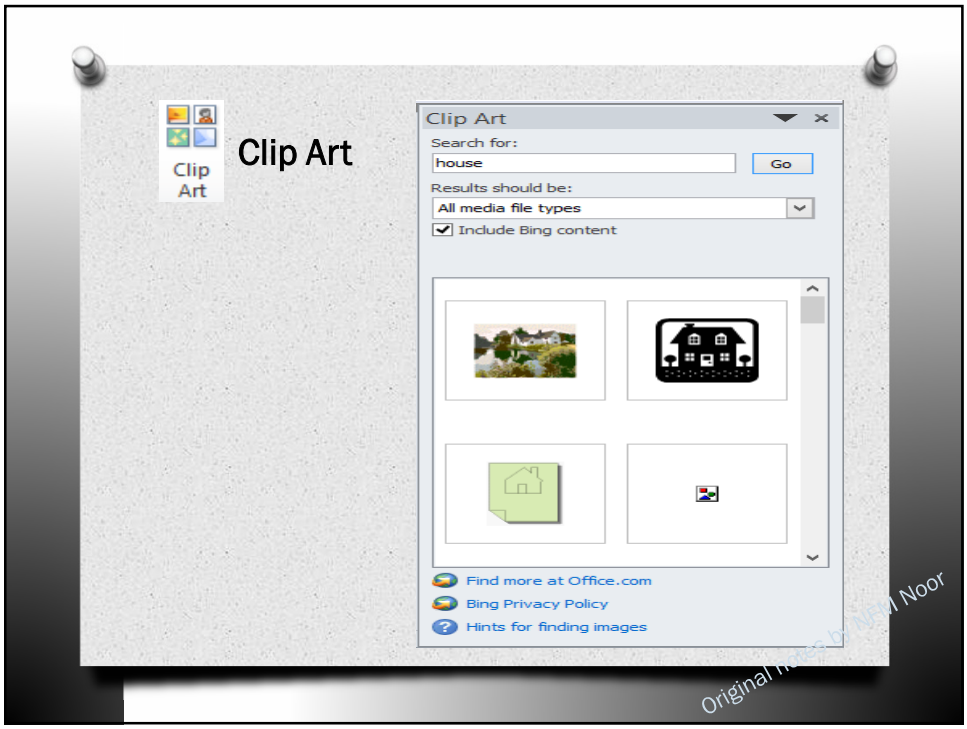
Graphics

- Picture
- Clip Art
- Shapes
- Smart Art
- Screenshot

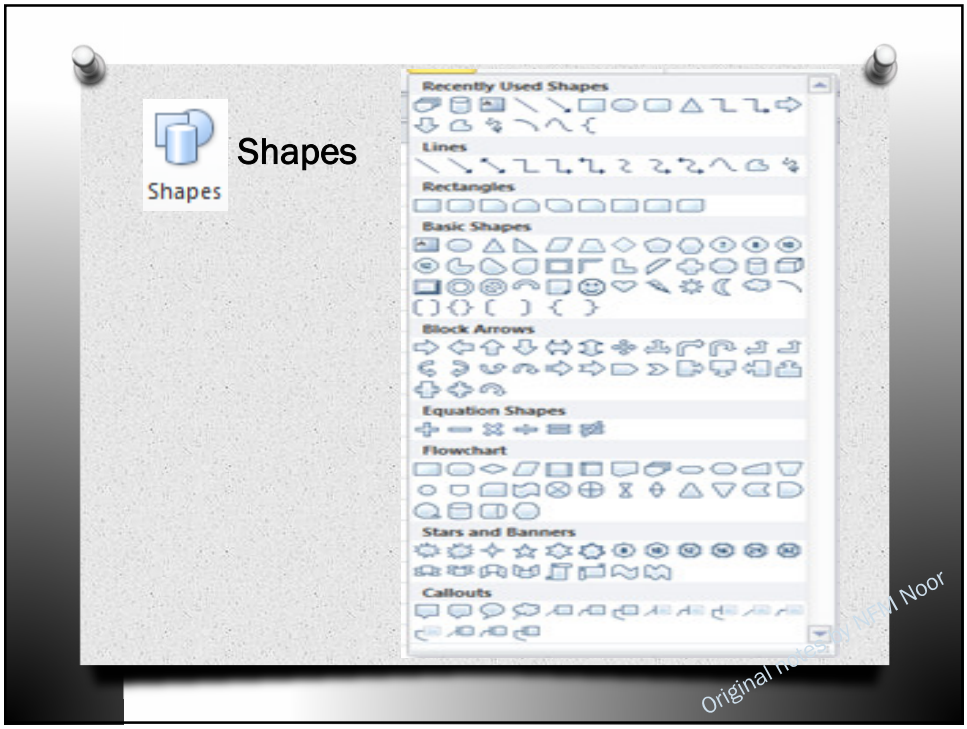


Original notes by NFM Noor

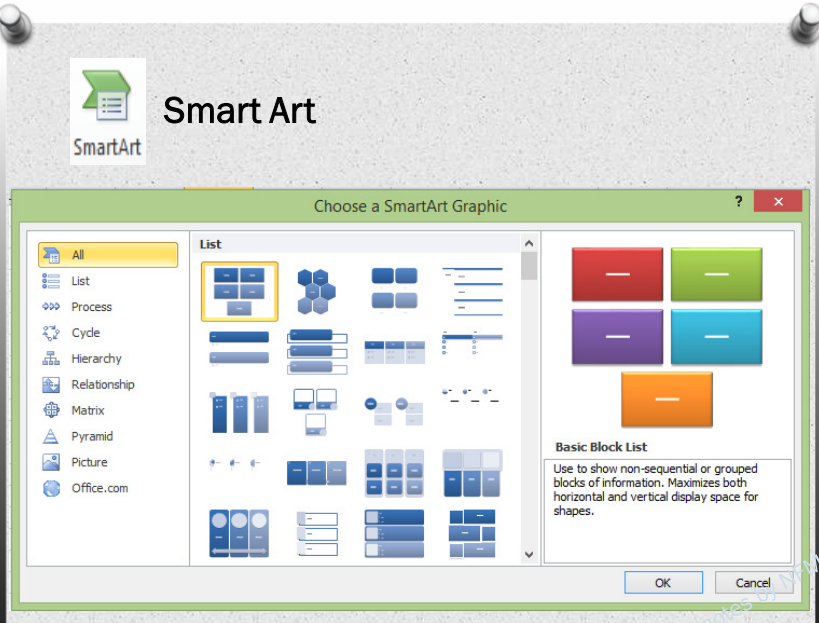
Clip Art



Shapes

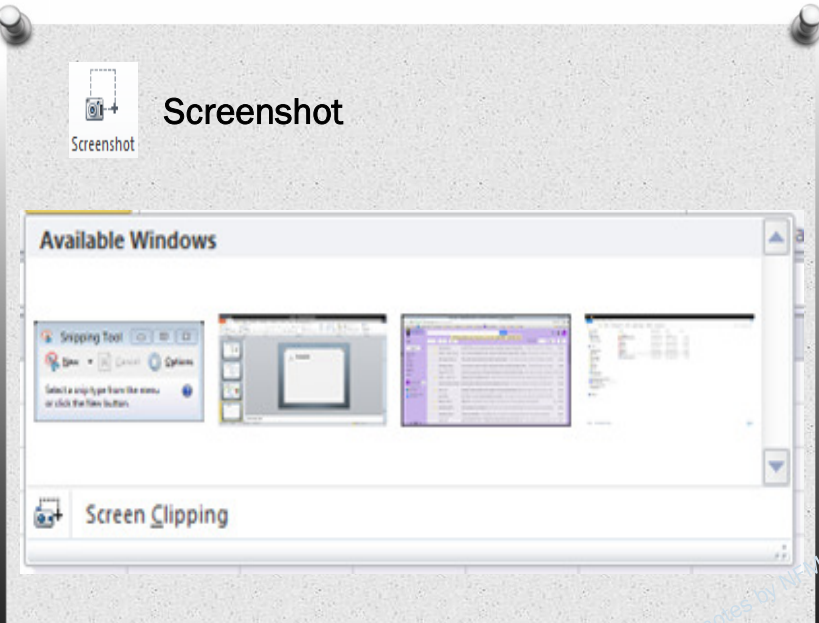


Smart Art



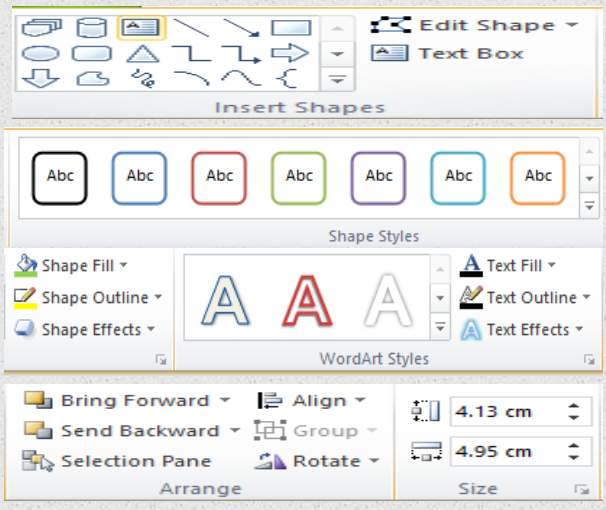
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Screenshot



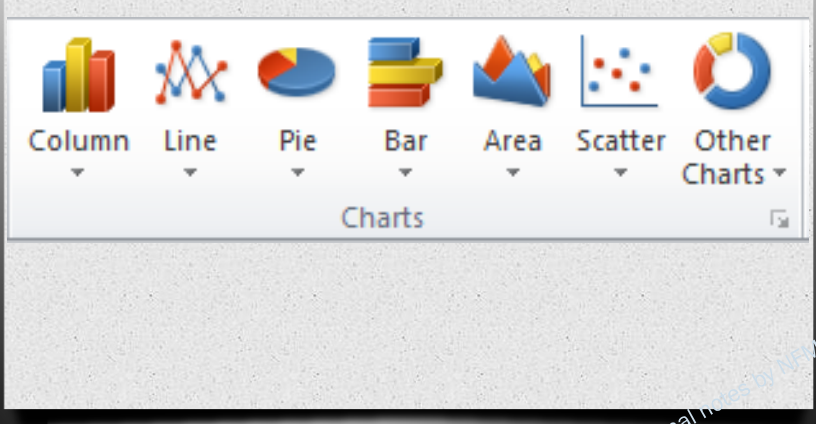
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Drawing Tools

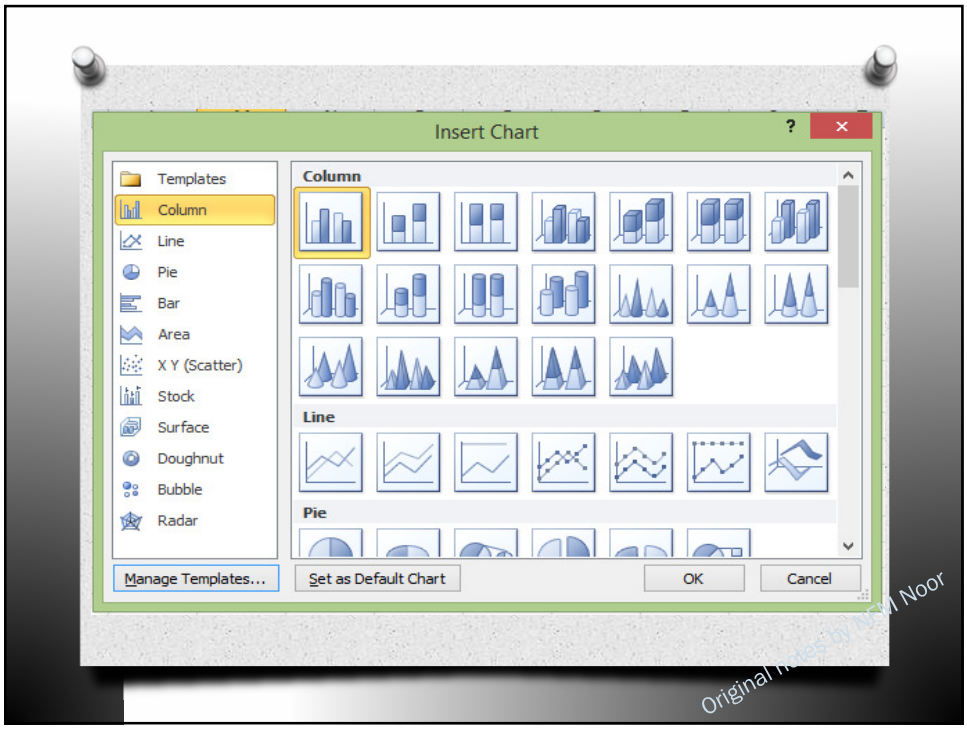


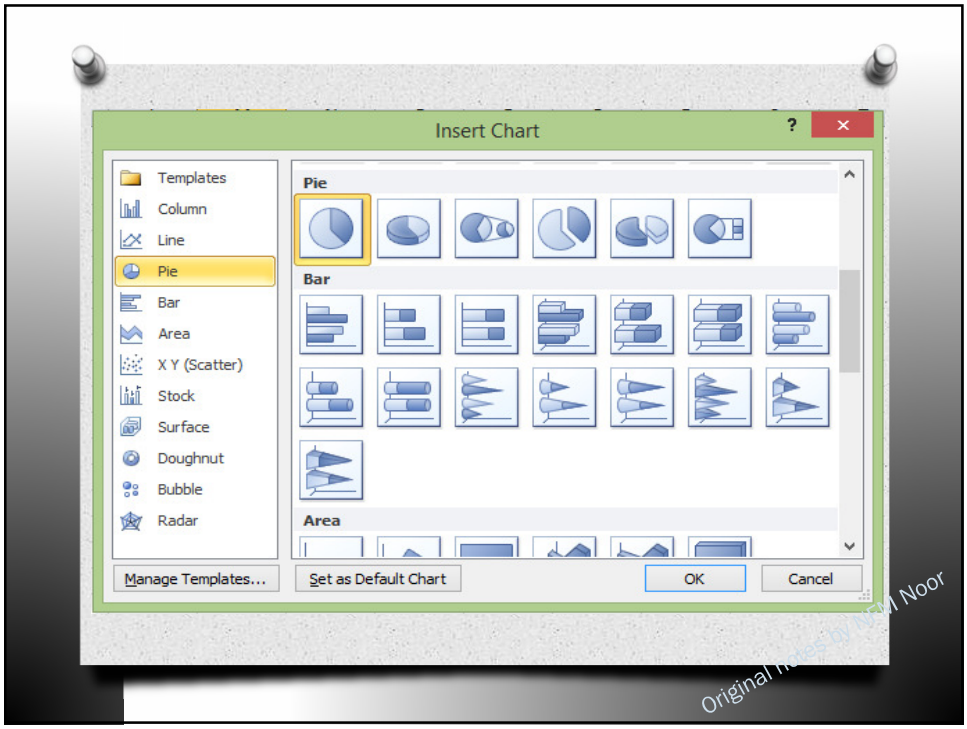
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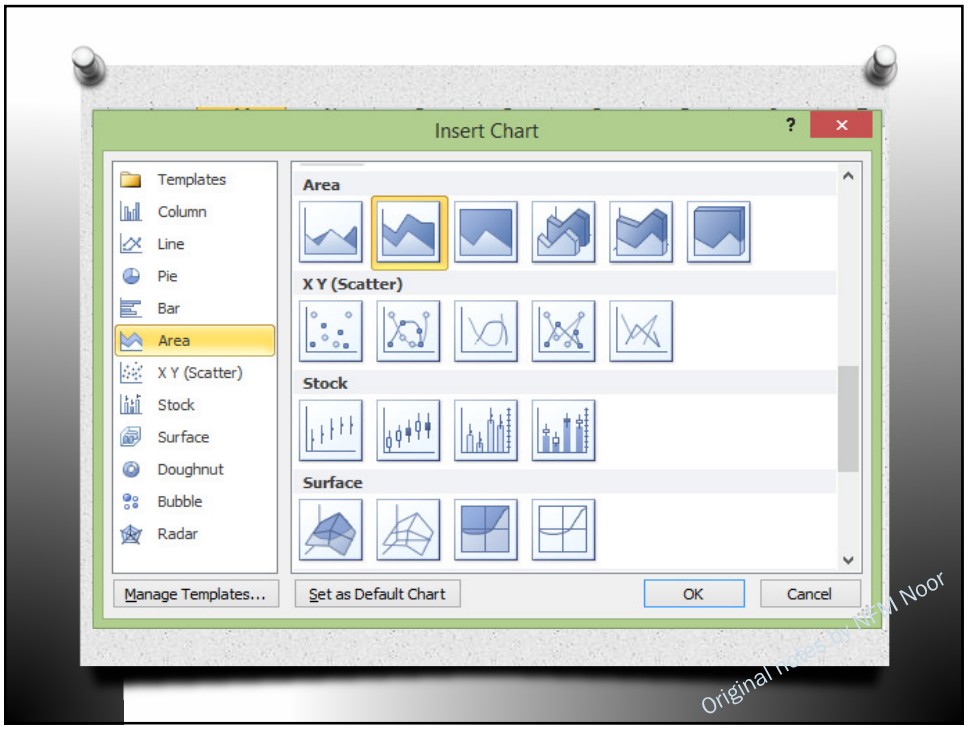
Charts

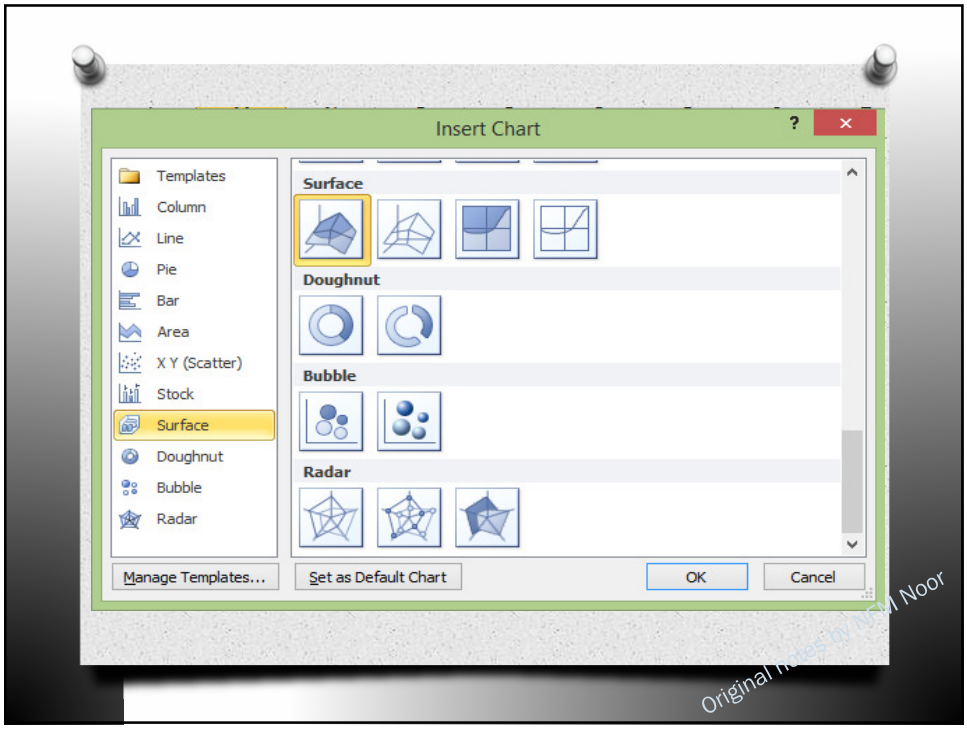


Original notes by NFM Noor


Insert Chart
Templates
Column
Line
Pie
Bar
Area
X Y (Scatter)
Stock
Surface
Doughnut
Bubble
Radar
Column
Line
Pie
Manage Templates...
Set as Default Chart
OK
Cancel
Original notes by NFM Noor


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Set as Default Chart
OK
Cancel
Original notes by NFM Noor

Creating A Chart

Step 1: Create a basic chart

Step 2: Change the layout or style of a chart

Step 3: Add or remove titles or data labels

Step 4: Show or hide a legend

Step 5: Display or hide chart axes or gridlines

Step 6: Move or resize a chart

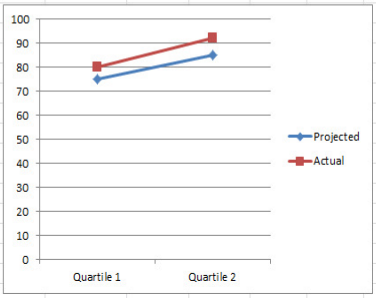
Step 7: Save a chart as a template

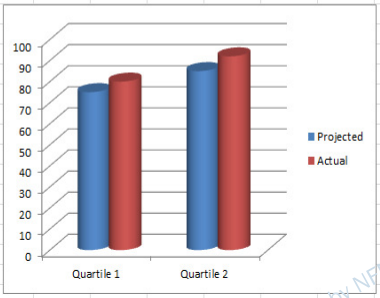
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Example

	Quartile 1	Quartile 2
Projected	75	85
Actual	80	92

1. Create table of data.
2. Insert chart.
3. On the blank chart area, right click and "Select Data".
4. Use Cursor to highlight all cells and press "OK".



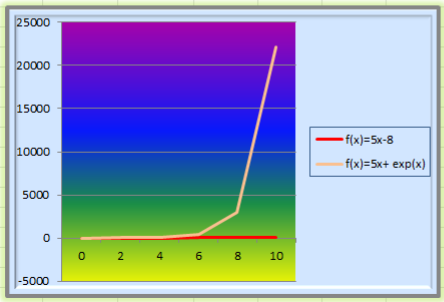


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Example

x	f(x)=5x-8	f(x)=5x+ exp(x)
0	-8	1
2	2	17.3890561
4	12	74.59815
6	22	433.428793
8	32	3020.95799
10	42	22076.4658

1. Create table of data using function.
2. Insert chart.
3. On the blank chart area, right click and "Select Data".
4. Use Cursor to highlight all cells.
5. Edit, Add and Remove accordingly and press "OK".



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Curve Fitting

Process of constructing a curve (or mathematical function) that best fits a series of data points.

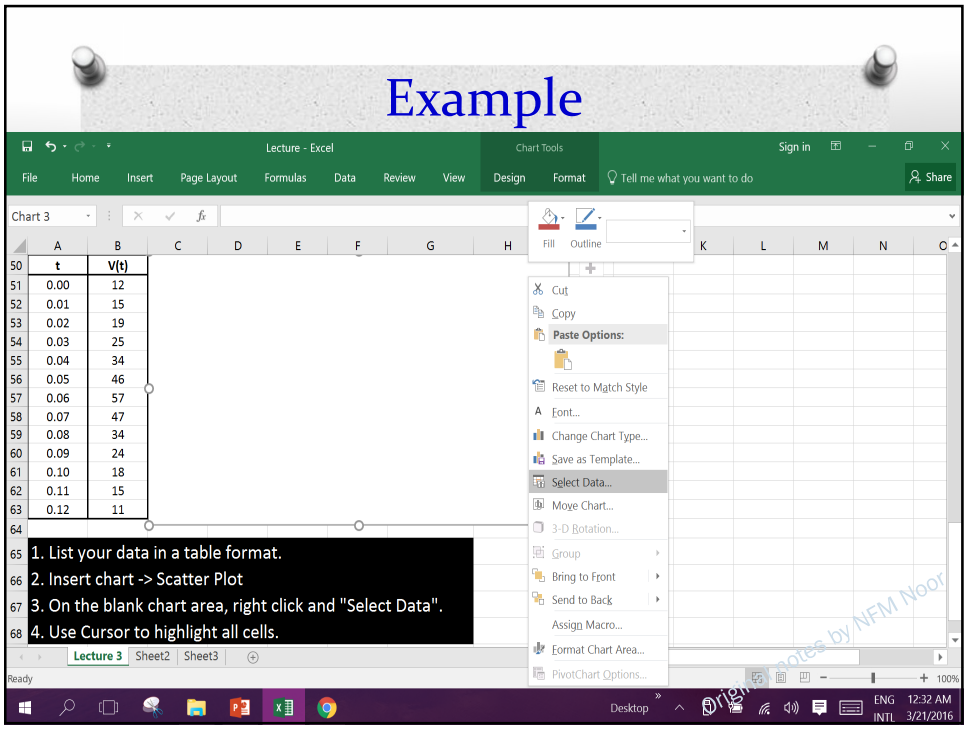
It involves finding the relationship between an independent variable (x) and a dependent variable (y) by minimizing the difference between the observed data and the values predicted by the model.

Types of models/functions available:

- Linear
- Polynomial
- Power
- Exponential
- Logarithmic
- Moving Average

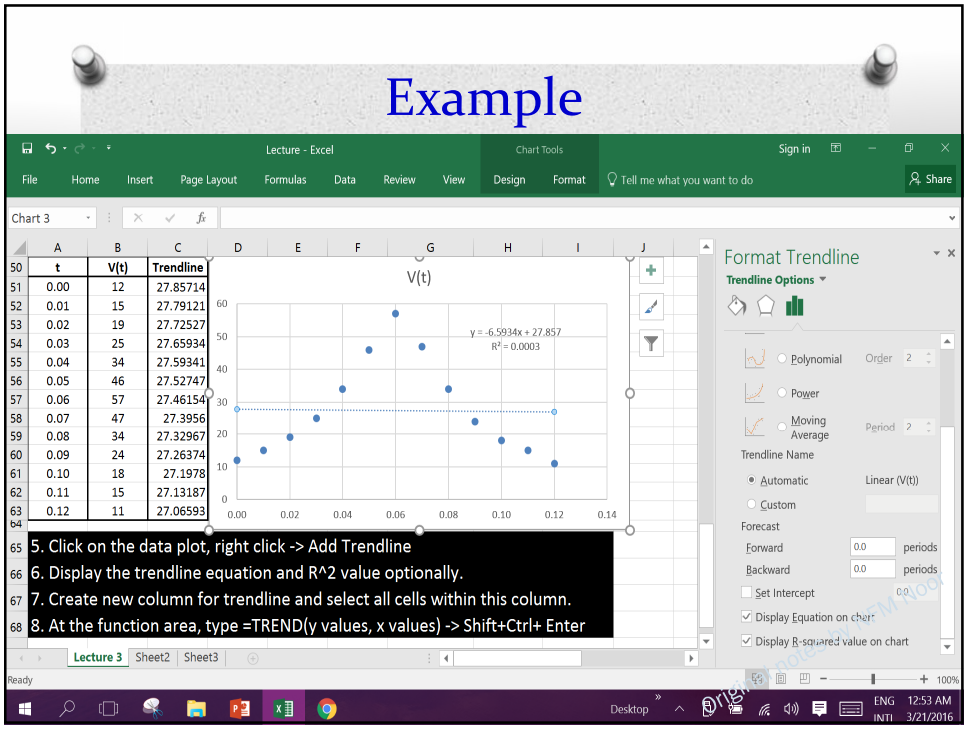
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Example



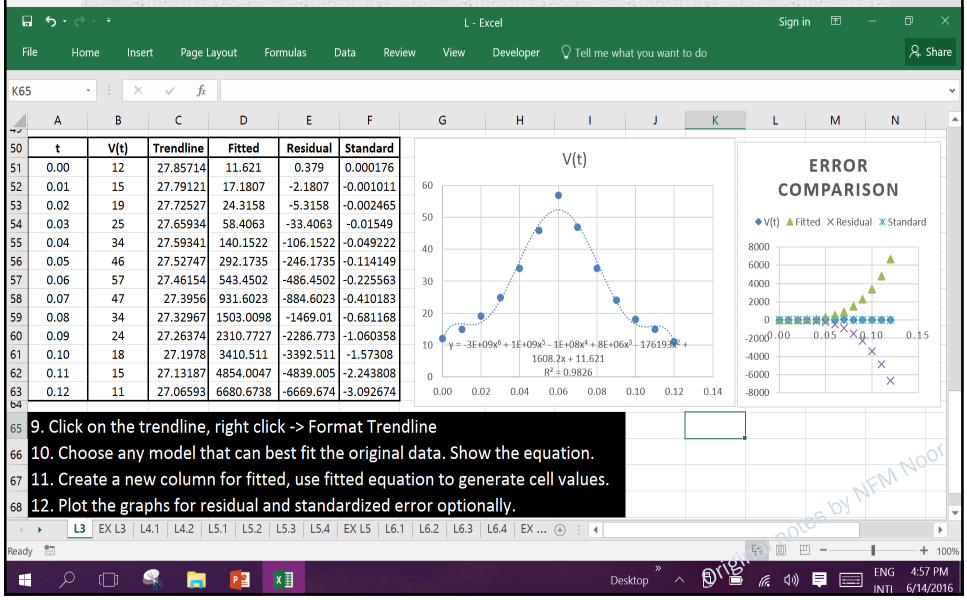
t	V(t)
0.00	12
0.01	15
0.02	19
0.03	25
0.04	34
0.05	46
0.06	57
0.07	47
0.08	34
0.09	24
0.10	18
0.11	15
0.12	11

Example



t	V(t)	Trendline
0.00	12	27.85714
0.01	15	27.79121
0.02	19	27.72527
0.03	25	27.65934
0.04	34	27.59341
0.05	46	27.52747
0.06	57	27.46154
0.07	47	27.3956
0.08	34	27.32967
0.09	24	27.26374
0.10	18	27.1978
0.11	15	27.13187
0.12	11	27.06593

Example



t	V(t)	Trendline	Fitted	Residual	Standard
0.00	12	27.85714	11.621	0.379	0.000176
0.01	15	27.79121	17.1807	-2.1807	-0.001011
0.02	19	27.72527	24.3158	-5.3158	-0.002465
0.03	25	27.65934	58.4063	-33.4063	-0.01549
0.04	34	27.59341	140.1522	-106.1522	-0.049222
0.05	46	27.52747	292.1735	-246.1735	-0.114149
0.06	57	27.46154	543.4502	-486.4502	-0.225563
0.07	47	27.3956	931.6023	-884.6023	-0.410183
0.08	34	27.32967	1503.0098	-1469.01	-0.681168
0.09	24	27.26374	2310.7727	-2286.773	-1.060358
0.10	18	27.1978	3410.511	-3392.511	-1.57308
0.11	15	27.13187	4854.0047	-4839.005	-2.243808
0.12	11	27.06593	6680.6738	-6669.674	-3.092674

Summary

- Excel provides graphical tools for data analysis and presentation.
- A chart can be created from a list of data prepared in a worksheet.
- Excel provides curve fitting tools to enable a user to conduct trend analysis, data smoothing, interpolation and extrapolation.

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